

MICHALIS CHADOLIAS

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Doctoral student in neutrino physics at the Instituto de Física Corpuscular (IFIC) and the Universitat de València, working within the KM3NeT collaboration. Research experience spans neutrino physics analyses, simulation and data-processing pipelines, and machine-learning methods in astroparticle physics.

RESEARCH INTERESTS

Neutrino physics, Particle & astroparticle physics, Statistical & Computational methods, Data science & AI

EDUCATION

Ph.D. in Neutrino Physics <i>Universitat de València (UV) / Instituto de Física Corpuscular (IFIC)</i>	Feb 2026 – Feb 2030 <i>Valencia, Spain</i>
M.Sc. in Astroparticle Physics <i>Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU); Grade: 1.6/1.0</i>	Apr 2022 – Feb 2025 <i>Erlangen, Germany</i>
B.Sc. in Physics <i>Aristotle University of Thessaloniki (AUTH); Grade: 7.84/10.0</i>	Oct 2017 – Feb 2022 <i>Thessaloniki, Greece</i>

RESEARCH EXPERIENCE

Predoctoral Researcher <i>Instituto de Física Corpuscular (IFIC) - VEGA Group</i>	Feb 2026 – Present <i>Valencia, Spain – Full-time</i>
- Planned contribution to the Data/MC comparison efforts for the ORCA23/24 data, focusing on event classification and spark rejection for background reduction through BDTs - Position funded through the FPI fellowship from the Spanish Ministry of Science and Innovation.	
Master Thesis Student <i>Erlangen Centre for Astroparticle Physics (ECAP) Thesis</i>	Aug 2023 – Oct 2024 <i>Erlangen, Germany – Full-time</i>
- Conducted the first feasibility study on tau neutrino appearance using ANTARES simulated data, evaluating the reconstruction quality of the events and providing the first estimate of the detector's capabilities. - Developed a scalable pipeline to convert over 500k ANTARES Monte Carlo event files to TTree-based ROOT files, reducing pre-processing time by 20%, storage requirements by 35%, and enabling parallelised cluster execution (NHR@FAU & CC@Lyon). - Analysed 200+ different oscillation scenarios with the SWIM software, accounting for the possible cut selection criteria, model parameters, and systematic uncertainties. - Compared the tau appearance potential from ANTARES against KM3NeT/ORCA predicted sensitivities.	
Research Assistant (HiWi) <i>Erlangen Centre for Astroparticle Physics (ECAP)</i>	Nov 2022 – Apr 2023 <i>Erlangen, Germany – Contract</i>
- Co-developed and tested modules within a Snakemake-based MC production pipeline for ORCA events. - Performed testing and diagnostics of pipeline jobs on NHR@FAU cluster, reducing execution errors and improving workflow reproducibility. - Achieved 25% CPU workload reduction by improving the SLURM profile, identifying inefficient parallelisation patterns, and optimising Snakemake rule resources.	
Undergraduate Research Assistant <i>National Centre of Scientific Research Demokritos (NCSR) Report</i>	Nov 2021 – Jan 2022 <i>Athens, Greece – Internship</i>
- Studied the reconstruction quality of simulated events focusing on the limited 6-string KM3NET/ORCA detector configuration (ORCA6). - Analysed ν_μ interaction topologies across key metrics such as energy, zenith angle, inelasticity, and distance from the detector, among others. - Assisted in the optical module integration and component testing.	
Undergraduate Researcher & Sub-team Coordinator <i>SpaceDot Team - AcubeSAT Project Thesis</i>	Mar 2019 – Dec 2021 <i>Thessaloniki, Greece – Part-time</i>
- Led the trajectory subsystem and co-authored a proposal for ESA's "Fly Your Satellite" programme. - Modelled the space environment for the mission with the OMERE, and estimated the radiation levels during the entire duration of the mission with the FASTRAD software, with this work being credited as my bachelor's thesis.	

FUNDING

FPI Predoctoral Fellowship (Formación de Profesorado Investigador)

Spanish Ministry of Science and Innovation

Issued on: Jan 2026

Valencia, Spain

Internship Mobility Grant

Aristotle University of Thessaloniki (AUTH)

Issued on: Oct 2021

Thessaloniki, Greece

SKILLS

Languages: Python, C/C++, Bash, SQL

Frameworks & Tools: ROOT, SWIM, OscProb, PyTorch, GraphNet, TensorFlow, nuSQuIDS

Computing Methods: High-Performance Computing (SLURM, UNIX), Workflow Management (Snakemake), Version Control (Git), CI/CD (GitHub Actions), Containerization (Docker, Apptainer)

Foreign Languages: Greek (Native), English (C2), German (B2), Italian (A1)

Soft Skills: Team player, analytical thinker, problem-solving, adaptability

CONFERENCES & WORKSHOPS

ANTARES-KM3NeT Analysis Meeting

Participant (Remote)

Oct 2025

CERN, CH

Analysis Reproducibility

Participant

Mar 2025

Online

12th HEP C++ Course and Hands-on Training - The Essentials

Participant

Mar 2025

Online

KM3NeT Bootcamp

Participant

Dec 2024

Erlangen, DE

DPG Spring Meeting

Speaker: Feasibility study of tau appearance measurement with the ANTARES neutrino telescope (T 29.2)

Mar 2024

Karlsruhe, DE

KM3NeT Oscillation WG Meeting

Speaker: Feasibility study of tau appearance measurement with the ANTARES neutrino telescope

Dec 2023

Erlangen, DE

KM3NeT Bootcamp

Participant (Remote)

Oct 2023

Paris, FR

GraphNeT III - Workshop

Participant

May 2023

Sandvig, DK

VOLUNTEERING

Long Night of Sciences, FAU

Volunteer - Neutrino Group ECAP

Erlangen, Germany

October 2023

- Presented neutrino detection concepts to 1100+ visitors at ECAP through posters and demonstrations, as well as showcasing the group's contributions to neutrino physics & astronomy.

AUTH - CERN Masterclass

Volunteer

Thessaloniki, Greece

Mar 2021

- Guided 10 high school students through ATLAS event identification exercises during the AUTH-CERN Masterclass.

Physicists of Aristotle University of Thessaloniki (PATH)

Group Coordinator

Thessaloniki, Greece

Nov 2017 – Jun 2019

- Organized the speaker list and conference logistics to highlight undergraduate research in physics for the second iteration of the $\Sigma\Pi E\Phi$ conference.
- Facilitated student visits to research labs through the 'Labs & Paths' event, helping fellow students engage with the research conducted in the department.

PUBLICATIONS

1. Aiello, S. *et al.* gSeaGen code by KM3NeT: An efficient tool to propagate muons simulated with CORSIKA. *Comput. Phys. Commun.* **314**, 109660. arXiv: 2410.24115 [hep-ex] (2025).
2. Aiello, S. *et al.* Measurement of neutrino oscillation parameters with the first six detection units of KM3NeT/ORCA. *JHEP* **10**, 206. arXiv: 2408.07015 [hep-ex] (2024).
3. Aiello, S. *et al.* Search for neutrino emission from GRB 221009A using the KM3NeT ARCA and ORCA detectors. *JCAP* **08**, 006. arXiv: 2404.05354 [astro-ph.HE] (2024).
4. Aiello, S. *et al.* Atmospheric muons measured with the KM3NeT detectors in comparison with updated numeric predictions. *Eur. Phys. J. C* **84**, 696. arXiv: 2403.11946 [astro-ph.HE] (2024).
5. Aiello, S. *et al.* Differential Sensitivity of the KM3NeT/ARCA detector to a diffuse neutrino flux and to point-like source emission: Exploring the case of the Starburst Galaxies. *Astropart. Phys.* **162**, 102990. arXiv: 2402.09088 [astro-ph.HE] (2024).
6. Aiello, S. *et al.* Astronomy potential of KM3NeT/ARCA. *Eur. Phys. J. C* **84**, 885. arXiv: 2402.08363 [astro-ph.HE] (2024).
7. Aiello, S. *et al.* Searches for neutrino counterparts of gravitational waves from the LIGO/Virgo third observing run with KM3NeT. *JCAP* **04**, 026. arXiv: 2311.03804 [astro-ph.HE] (2024).
8. Unbehaun, T. *et al.* Prospects for combined analyses of hadronic emission from γ -ray sources in the Milky Way with CTA and KM3NeT. *Eur. Phys. J. C* **84**, 112. arXiv: 2309.03007 [astro-ph.HE] (2024).
9. Aiello, S. *et al.* Embedded software of the KM3NeT central logic board. *Comput. Phys. Commun.* **296**, 109036. arXiv: 2308.01032 [astro-ph.IM] (2024).